Do Now · retrieval

Half adders and full adders

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

When we group four neighbouring 1s on a Karnaugh map, what happens to the Boolean expression?

Q2

Represent −13 using 8-bit sign and magnitude.

Q3

Insert 40, 20, 60, 10, 30, 50 into an empty binary search tree. Draw the resulting tree.

Q4

Using a 5-bit two’s complement mantissa and 4-bit two’s complement exponent, calculate A − B. Show exponent alignment and mantissa subtraction. A = $0.1100_2 \times 2^3$; B = $0.1000_2 \times 2^2$.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Start with binary addition — One binary column can produce a sum bit and a carry bit.

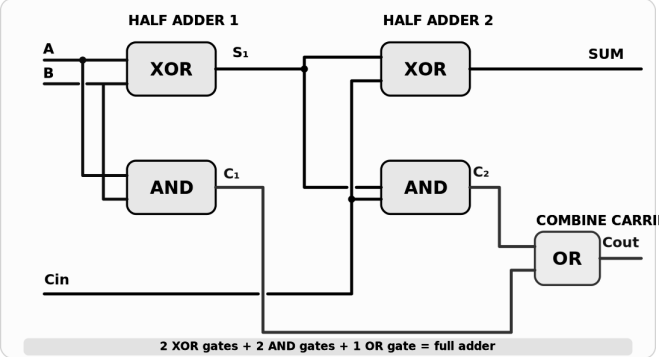
Core idea 2

Discover the two gates — XOR calculates Sum; AND calculates Carry.

Core idea 3

Find the limitation — What happens when a previous column sends a carry into this column?

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

First half adder — Add A and B to produce an intermediate Sum S_1 and Carry C_1.

Core idea 5

Combine both carries — Feed C_1 and C_2 into an OR gate to obtain Carry-out.

Vocabulary · use the correct technical terms

half adder — A circuit adding two input bits.

full adder — A circuit adding two bits plus a carry-in.

Sum — The result bit for the current column.

Carry — A bit passed to the next column.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Model 1 + 1 with a half adder

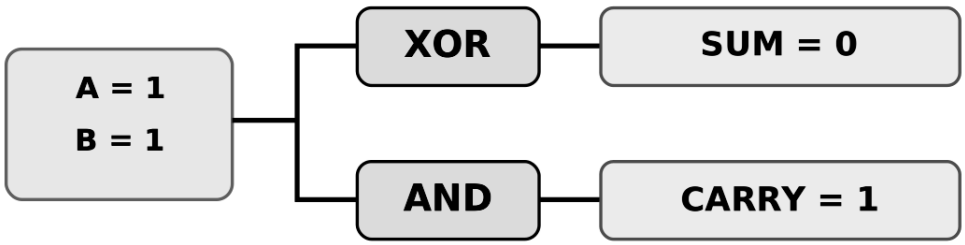
Follow the worked method

1. Rightmost result bit — CODE:
Sum = 0

2. Trace XOR and AND — CODE:
1 XOR 1 = 0 → Sum
1 AND 1 = 1 → Carry

3. Connect both half adders — CODE:
S_1 = A XOR B
Sum = S_1 XOR Cin
Cout = (A AND B) OR (S_1 AND Cin)

Source model · final illustrated step



1 + 1 = 10_2 → sum bit 0, carry bit 1

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Complete the same formulae
CODE:
S = A ? B
C = A ? B

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

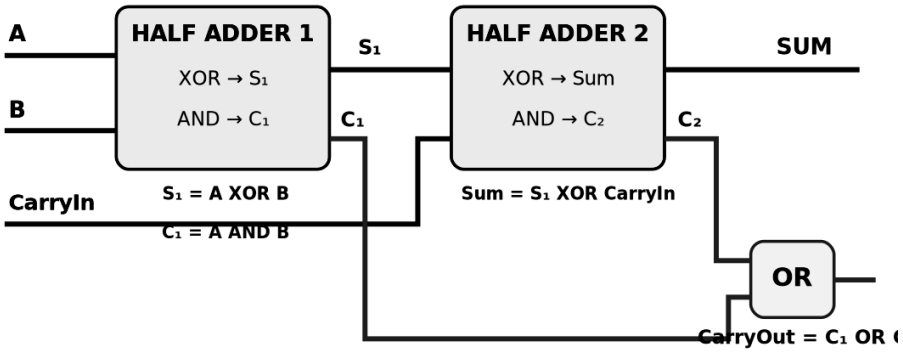
Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Half-adder question

State the Sum/Carry expressions, complete the truth table, then explain how half adders contribute to a full adder.

Question 1 · original source diagram



Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Add 0111 + 0001

Use bit-by-bit addition and identify where the carry travels.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

For a half adder, Sum is produced by XOR and Carry is produced by AND.

Use key vocabulary from memory

1. Explain half adder in your own words.

2. Explain full adder in your own words.

3. Explain Sum in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.